Notes on the Current RG-Tail Waveform Implementation

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1 Purpose

This note summarizes the current implementation in `src/rg_tail/waveform.py`. It replaces the earlier separate summary, paper-audit, and $h(f)$-prescription notes.

The goal is to describe, in one place:

- what the current code implements,
- which parts come directly from Section IV of arXiv:2602.08833,
- which parts are repository-level modelling choices,
- and what the main current caveats are.

2 High-Level Picture

The core paper-based object is the factorized $(2,2)$ mode correction

$$\hat{h}_{22} = \hat{S}^{(0)}_{\rm eff}\, T^{\rm rad}_{22}\, \rho^2_{22}\, e^{i\delta_{22}}, \tag{1}$$

which is the Section IV comparable-mass specialization of Eq. (131) once the internal-tail bracket is omitted at 4PN.

The current repository then turns this into a practical Fourier-domain balance-law template by using the dominant-mode flux implied by $\hat{h}_{22}$ inside the adiabatic balance law. The core flux model is

$$\mathcal{F}_{22}(x;\lambda_{\rm RG}) = \frac{32}{5}\nu^2 x^5 \left|\hat{h}_{22}(x;\lambda_{\rm RG})\right|^2, \tag{2}$$

and the Fourier-domain waveform is built from the corresponding stationary-phase phase evolution.

This distinction is important:

- $\hat{h}_{22}$ is the paper-facing factorized correction,
- $\mathcal{F}_{22}$ is the 22-only radiative flux implied by that correction,
- $\tilde{h}(f)$ is the repository's data-analysis wrapper built from this flux.

3 What `waveform.py` Contains

The module is organized in four blocks:

1. test-mass ingredients used for validation,
2. comparable-mass Section IV ingredients,
3. the clean factorized correction $\hat{h}_{22}$,
4. the 22-only balance-law flux and SPA phase helpers,
5. lightweight detector helpers.

3.1 Test-mass branch

The test-mass branch is kept mainly for reproducing the self-convergence checks discussed in the paper. It includes:

- exact Schwarzschild $E_{\rm circ}$ and $p_\phi^{\rm circ}$,
- the $\hat{\ell}_2$ expansion,
- test-mass ρ_{22}, δ_{22},
- test-mass $\tilde{f}_{22}$, $\tilde{\delta}_{22}$.

These are not the main comparable-mass forecast model, but they are useful for checking that the resummed structure behaves as expected.

3.2 Comparable-mass Section IV branch

For $\nu > 0$, the module implements:

- $E_{\rm real}/M$ from Eq. (145),
- $p_{\phi,{\rm circ}}/(\mu M)$ from Eq. (146),
- $\gamma_{22}^{\rm univ}$ from Eq. (137),
- the radiative tail factor $T_{22}^{\rm rad}$ from Eq. (155),
- ρ_{22} from Eq. (157),
- δ_{22} from Eq. (158).

The effective source is implemented through

$$\hat{H}_{\rm eff} = \frac{(E_{\rm real}/M)^2 - 1}{2\nu} + 1, \tag{3}$$

which is the Eq. (132) relation solved for the even-parity source.

4 Explicit Current Model Formulas

This section collects the main formulas used by the current code in a single place.

4.1 Basic variables

For the dominant $(2,2)$ mode, the code uses

$$x = (M\Omega)^{2/3}, \qquad \omega = 2\Omega, \qquad f = \frac{\omega}{2\pi} = \frac{\Omega}{\pi}, \qquad x_f = (\pi M f)^{2/3}. \tag{4}$$

The symmetric mass ratio is $\nu = \eta$.

4.2 Conservative circular-orbit sector

The current comparable-mass circular-orbit energy is

$$\frac{E_{\rm real}}{M} = 1 - \frac{\nu x}{2}\left(c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4\right), \tag{5}$$

with

$$c_0 = 1, \tag{6}$$

$$c_1 = -\frac{3}{4} - \frac{\nu}{12}, \tag{7}$$

$$c_2 = -\frac{27}{8} + \frac{19}{8}\nu - \frac{\nu^2}{24}, \tag{8}$$

$$c_3 = -\frac{675}{64} + \left(\frac{34445}{576} - \frac{205\pi^2}{96}\right)\nu - \frac{155}{96}\nu^2 - \frac{35}{5184}\nu^3, \tag{9}$$

$$c_4 = -\frac{3969}{128} + \left(-\frac{123671}{5760} + \frac{9037\pi^2}{1536} + \frac{896\gamma_E}{15} + \frac{448}{15}\log(16x)\right)\nu + \left(\frac{498449}{3456} - \frac{3157\pi^2}{576}\right)\nu^2 + \frac{301}{1728}\nu^3 + \frac{77}{31104}\nu^4. \tag{10}$$

The current circular angular momentum is

$$\frac{p_{\phi,\rm circ}}{\mu M} = \frac{1}{\sqrt{x}}\left(d_0 + d_1 x + d_2 x^2 + d_3 x^3 + d_4 x^4\right), \tag{11}$$

with

$$d_0 = 1, \tag{12}$$

$$d_1 = \frac{3}{2} + \frac{\nu}{6}, \tag{13}$$

$$d_2 = \frac{27}{8} - \frac{19}{8}\nu + \frac{\nu^2}{24}, \tag{14}$$

$$d_3 = \frac{135}{16} + \left(-\frac{6889}{144} + \frac{41\pi^2}{24}\right)\nu + \frac{31}{24}\nu^2 + \frac{7}{1296}\nu^3, \tag{15}$$

$$d_4 = \frac{2835}{128} + \left(\frac{98869}{5760} - \frac{128\gamma_E}{3} - \frac{6455\pi^2}{1536} - \frac{64}{3}\log(16x)\right)\nu + \left(\frac{356035}{3456} - \frac{2255\pi^2}{576}\right)\nu^2 - \frac{215}{1728}\nu^3 - \frac{55}{31104}\nu^4. \tag{16}$$

The effective source is

$$\hat{H}_{\rm eff} = \frac{(E_{\rm real}/M)^2 - 1}{2\nu} + 1. \tag{17}$$

4.3 RG-tail sector

The auxiliary quantities entering the universal anomalous dimension are

$$\hat{k} = \frac{E_{\rm real}}{M}\, m\Omega, \qquad J = \frac{p_{\phi,\rm circ}/(\mu M)}{(E_{\rm real}/M)^2}, \tag{18}$$

with $m = 2$ in the current model.

The universal anomalous dimension is

$$\gamma_{22}^{\rm univ} = -\frac{214}{105}\hat{k}^2 + \frac{2mJ}{3}\hat{k}^3 - \frac{3390466}{1157625}\hat{k}^4 + \frac{381863\, mJ}{99225}\hat{k}^5. \tag{19}$$

The repository-level RG deformation is

$$\hat{\hat{\ell}}_{22} = 2 + \lambda_{\rm RG}\, \gamma_{22}^{\rm univ}, \tag{20}$$

with GR corresponding to $\lambda_{\rm RG} = 1$.

The radiative-coordinate external tail factor is

$$T_{22} = \exp\Big[\log(120) + (\hat{\hat{\ell}}_{22} - 2)\log(2kr_\omega) + 2i\hat{k}\log(4\phi_0) + \log\Gamma(\hat{\hat{\ell}}_{22} - 1 - 2i\hat{k}) - \log\Gamma(2\hat{\hat{\ell}}_{22} + 2) + \pi\hat{k} - \frac{i\pi}{2}(\hat{\hat{\ell}}_{22} - 2)\Big], \tag{21}$$

where

$$k = 2\Omega, \qquad r_\omega = \frac{1}{x}, \qquad \phi_0 = \frac{e^{17/12 - \gamma_E}}{4}. \tag{22}$$

4.4 Residual amplitude and residual phase

Define

$$\mathrm{eulerlog}_2(x) = \gamma_E + \log(4\sqrt{x}). \tag{23}$$

Then the current comparable-mass residual amplitude is

$$\rho_{22}(x,\nu) = 1 + r_1 x + r_2 x^2 + r_3 x^3 + r_4 x^4, \tag{24}$$

with

$$r_1 = -\frac{43}{42} + \frac{55}{84}\nu, \tag{25}$$

$$r_2 = -\frac{20555}{10584} - \frac{33025}{21168}\nu + \frac{19583}{42336}\nu^2, \tag{26}$$

$$r_3 = -\frac{4296031}{4889808} + \left(\frac{41\pi^2}{192} - \frac{48993925}{9779616}\right)\nu - \frac{6292061}{3259872}\nu^2 + \frac{10620745}{39118464}\nu^3, \tag{27}$$

$$\begin{aligned} r_4 = {} & \frac{9228174993589}{800950550400} + \left(-\frac{2487107795131}{145627372800} + \frac{464}{35}\mathrm{eulerlog}_2(x) - \frac{9953\pi^2}{21504}\right)\nu \\ & + \left(\frac{10815863492353}{640760440320} - \frac{3485\pi^2}{5376}\right)\nu^2 - \frac{2088847783}{11650189824}\nu^3 + \frac{70134663541}{512608352256}\nu^4. \end{aligned} \tag{28}$$

The current residual phase is

$$\delta_{22} = -\frac{17}{3}y^{3/2} - 24\nu\, y^{5/2} + \left(\frac{30995}{1134}\nu + \frac{962}{135}\nu^2\right) y^{7/2} - \frac{4976\pi}{105}\nu\, y^4, \tag{29}$$

where

$$y = \left(\frac{E_{\rm real}}{M} x^{3/2}\right)^{2/3}. \tag{30}$$

The final Section IV-inspired factorized correction is therefore

$$\hat{h}_{22} = \hat{H}_{\rm eff}\, T_{22}\, \rho_{22}^2\, e^{i\delta_{22}}. \tag{31}$$

4.5 Flux, balance law, and SPA phase

Using the Newtonian mode

$$h_{22}^N = -8\sqrt{\frac{\pi}{5}}\,\nu x\, e^{-2i\phi}, \tag{32}$$

the current 22-only flux is

$$\mathcal{F}_{22}(x;\lambda_{\rm RG}) = \frac{32}{5}\nu^2 x^5 \left|\hat{h}_{22}(x;\lambda_{\rm RG})\right|^2. \tag{33}$$

The code uses the balance law

$$\frac{dt}{dx} = -M\,\frac{d(E_{\rm real}/M)/dx}{\mathcal{F}_{22}(x)}, \tag{34}$$

with

$$\frac{d\Phi_{\rm GW}}{dx} = 2\Omega\frac{dt}{dx} = 2\frac{x^{3/2}}{M}\frac{dt}{dx}. \tag{35}$$

If $x_{\rm ref}$ is the reference endpoint used to anchor the integrals, the code constructs

$$\Delta t(x) = \int_x^{x_{\rm ref}} \frac{dt}{dx'}\,dx', \qquad \Delta\Phi_{\rm GW}(x) = \int_x^{x_{\rm ref}} \frac{d\Phi_{\rm GW}}{dx'}\,dx', \tag{36}$$

and then uses the SPA phase

$$\Psi_{22,\rm bal}(f;\lambda_{\rm RG}) = \Delta\Phi_{\rm GW}(x_f) - 2\pi f\,\Delta t(x_f). \tag{37}$$

The current default Fourier-domain waveform uses the same balance law for the SPA amplitude. The Newtonian restricted amplitude is

$$A_N(f) = \sqrt{\frac{5}{24}}\,\frac{\mathcal{M}^{5/6}}{d_L\,\pi^{2/3}}\,f^{-7/6}. \tag{38}$$

Because the time-domain mode amplitude contains $|\hat{h}_{22}|$, while the inspiral rate contains $\mathcal{F}_{22} \propto |\hat{h}_{22}|^2$, the explicit $|\hat{h}_{22}|$ cancels in the stationary-phase amplitude:

$$|\hat{h}_{22}|\sqrt{\frac{\dot{f}_N}{\dot{f}}} = \sqrt{\frac{d(E_{\rm real}/M)/dx}{d(E_N/M)/dx}} = \sqrt{-\frac{2}{\nu}\frac{d(E_{\rm real}/M)}{dx}}. \tag{39}$$

Thus the default detector-domain template is

$$\tilde{h}(f) = A_N(f)\sqrt{-\frac{2}{\nu}\frac{d(E_{\rm real}/M)}{dx}\bigg|_{x_f}}\;\exp\!\left[i\left(2\pi f t_c - \phi_c - \frac{\pi}{4} + \Psi_{22,\rm bal}(f;\lambda_{\rm RG}) + \arg\hat{h}_{22}(x_f)\right)\right], \tag{40}$$

with

$$x_f = (\pi M f)^{2/3}. \tag{41}$$

The older project proxy, $\tilde{h}_{\rm proxy} = A_N|\hat{h}_{22}|\exp(i\Psi)$, is retained in the code as `amplitude_model="proxy"` for controlled comparisons.

4.6 Code-level `h_of_f` prescription

The current public waveform entry point is `h_of_f(f, Mc, eta, dL, tc, phic, lambda_RG, ...)`. In terms of the physical masses, the code first converts

$$M = \mathcal{M}_c\, \nu^{-3/5}, \qquad d_L = d_L^{\mathrm{Mpc}}\, \mathrm{Mpc}, \qquad M = M^{M_\odot}\, M_\odot, \tag{42}$$

with $G = c = 1$. It then defines the Schwarzschild ISCO frequency

$$f_{\mathrm{ISCO}} = \frac{1}{6^{3/2}\pi M}, \qquad f_{\mathrm{cut}} = \alpha_{\mathrm{cut}}\, f_{\mathrm{ISCO}}, \tag{43}$$

where $\alpha_{\mathrm{cut}} = \texttt{fmax\_over\_fisco}$. The waveform is set to zero outside the code band,

$$\tilde{h}(f) = 0, \qquad f < f_{\mathrm{low}} \quad \text{or} \quad f \geq f_{\mathrm{cut}}. \tag{44}$$

Inside the band, the exact current default is

$$\boxed{\tilde{h}(f) = W(f)\, A_{\mathrm{bal}}(f)\, \exp[i\Psi_{\mathrm{code}}(f)]}, \tag{45}$$

where the Fermi taper is

$$W(f) = \frac{1}{1 + \exp\left[\frac{f - f_{\mathrm{ISCO}}}{\sigma_{\mathrm{taper}} f_{\mathrm{ISCO}}}\right]}, \qquad \sigma_{\mathrm{taper}} = \texttt{sigma\_taper\_over\_fisco}. \tag{46}$$

The code phase is

$$\Psi_{\mathrm{code}}(f) = 2\pi f t_c - \phi_c - \frac{\pi}{4} + \Psi_{22,\mathrm{bal}}(f; \lambda_{\mathrm{RG}}) + \arg \hat{h}_{22}(x_f; \lambda_{\mathrm{RG}}), \tag{47}$$

with $x_f = (\pi M f)^{2/3}$. The default amplitude, `amplitude_model="balance"`, is

$$A_{\mathrm{bal}}(f) = A_N(f) \sqrt{\max\left[-\frac{2}{\nu} \frac{d(E_{\mathrm{real}}/M)}{dx}\bigg|_{x_f}, 0 \right]}. \tag{48}$$

The $\max[\cdots, 0]$ is a numerical guard in the code. For the current fiducial binaries and paper-facing frequency band, the physical branch has $d(E_{\mathrm{real}}/M)/dx < 0$.

The optional amplitude choices are

$$A_{\mathrm{proxy}}(f; \lambda_{\mathrm{RG}}) = A_N(f)\, \left| \hat{h}_{22}(x_f; \lambda_{\mathrm{RG}}) \right|, \tag{49}$$

$$A_{\mathrm{proxy,phase\ only}}(f) = A_N(f)\, \left| \hat{h}_{22}(x_f; \lambda_{\mathrm{RG}} = 1) \right|, \tag{50}$$

$$A_{\mathrm{newtonian}}(f) = A_N(f). \tag{51}$$

Thus `phase_only=True` only changes the old proxy amplitude; it does not change the current default balance-law amplitude, whose explicit λ_{RG}-dependent magnitude cancels in the SPA construction.

5 Current Relation to arXiv:2602.08833

Paper item	**Code function**	**Status**	**Current note**
Eqs. (85), (86)	`E_circ_test_mass`, `p_phi_circ_test_mass`	Match	Implemented in exact Schwarzschild form.
Eq. (175)	`ellhat_test_mass`	Match	The explicit test-mass $\hat{\ell}_2$ coefficients are kept.
Eq. (112)	`rho_22_test_mass`	Match	The current code now keeps the test-mass series through x^{10}.
Eq. (113)	`ftilde_22_test_mass`	Match	The current code now keeps the test-mass series through x^{10}.
Eqs. (114), (115)	`delta_22_test_mass`, `deltatilde_22_test_mass`	Match	Implemented as written in the paper.
Eq. (145)	`energy_real_over_M`	Match	Implemented at 4PN.
Eq. (146)	`p_phi_circ_over_mM`	Match	Implemented at 4PN.
Eq. (137)	`gamma_univ_22`	Match	Implemented with the paper's $\hat{k}$ and J definitions.
Eq. (155)	`tail_factor_T22`	Match up to intended 4PN use	The external radiative tail is implemented in log-space for numerical stability.
Eq. (157)	`rho_22`	Match	Implemented at 4PN.
Eq. (158)	`delta_22`	Match	Implemented at 4PN.
Eq. (131), 4PN Section IV use	`hhat_22`	Match to intended 4PN model	The external-tail-only 4PN form is used, consistent with the Section IV discussion below Eq. (143).
Internal-tail bracket in Eq. (131)	`include_internal_tail` option	Not implemented	The code now raises `NotImplementedError` instead of silently pretending to include it.
22-only flux $\mathcal{F}_{22}$	`flux_22`	Repository choice consistent with standard multipolar flux construction	Built from $h_{22} = h_{22}^N \hat{h}_{22}$ as $\mathcal{F}_{22} = (32/5)\nu^2 x^5 \lvert\hat{h}_{22}\rvert^2$.
Balance-law phase	`_spa_phase_from_22_flux`	Repository choice	The paper does not give a final SPA detector template, so the phase is built by integrating the 22-only flux with the GR conservative energy.
Detector-template $\tilde{h}(f)$	`h_of_f`	Repository choice	The final Fourier-domain waveform uses a balance-law SPA phase and, by default, the SPA amplitude correction implied by the same balance law. The older $A_N\lvert\hat{h}_{22}\rvert$ proxy is still available as an option.

6 Comparison with the 4PN Standard H_{22} Mode

The current model is also useful to compare against the standard 4PN dominant-mode result in arXiv:2304.11185. In that paper, Eq. (11) gives the complex $(2,2)$ mode H_{22} through relative 4PN order, using the normalization

$$h_+ - ih_\times = \frac{8Gm\nu x}{Rc^2}\sqrt{\frac{\pi}{5}}\sum_{\ell m} H_{\ell m} e^{-im\psi}\, {}_{-2}Y_{\ell m}. \tag{52}$$

With this normalization, the corresponding 22-mode flux is

$$\mathcal{F}_{22}^{2304.11185} = \frac{32}{5}\nu^2 x^5 |H_{22}|^2. \tag{53}$$

For $\lambda_{\rm RG} = 1$, the current code agrees with Eq. (11) through relative 3.5PN order. The first difference appears at relative 4PN:

$$H_{22}^{2304.11185} - H_{22}^{\rm code} = \frac{16}{45}(\nu - 1)\left[30\,\mathrm{eulerlog}_2(x) - 137 - 15i\pi\right] x^4, \tag{54}$$

where

$$\mathrm{eulerlog}_2(x) = \gamma_E + \ln(4\sqrt{x}). \tag{55}$$

Equivalently, at the level of the 22-mode flux,

$$\frac{\mathcal{F}_{22}^{2304.11185} - \mathcal{F}_{22}^{\rm code}}{(32/5)\nu^2 x^5} = \frac{32}{45}(\nu - 1)\left[30\,\mathrm{eulerlog}_2(x) - 137\right] x^4. \tag{56}$$

This term has the logarithmic plus $i\pi$ structure characteristic of hereditary radiative propagation effects. The companion derivation paper, arXiv:2304.11186, identifies a new 4PN "tail-of-memory" contribution in the radiative quadrupole moment, along with related tail-looking interactions. The current code should therefore be read as the Section IV external-tail factorized model of arXiv:2602.08833, not as the exact standard 4PN H_{22} mode of arXiv:2304.11185.

For now we do not add this 4PN difference to the waveform. If exact agreement with the standard 4PN H_{22} mode is desired later, one can multiply the current $\hat{h}_{22}$ by

$$1 + \frac{16}{45}(\nu - 1)\left[30\,\mathrm{eulerlog}_2(x) - 137 - 15i\pi\right] x^4, \tag{57}$$

or equivalently add the corresponding real flux correction above.

7 Current Beyond-GR Convention

The parameter $\lambda_{\rm RG}$ is not part of the GR waveform in the paper. It is a repository-level deformation used for Fisher forecasts. The current convention is

$$\hat{\hat{\ell}}_{22} = 2 + \lambda_{\rm RG}\, \gamma_{22}^{\rm univ}, \tag{58}$$

with GR corresponding to $\lambda_{\rm RG} = 1$.

The same scaling is applied in the test-mass branch by rescaling the running part $\hat{\ell}_2 - 2$.

8 Current Flux and Fourier-Domain Prescription

The paper gives the radiative mode structure and the harmonic relation $\omega = m\Omega$, but it does not give a final detector-template stationary-phase waveform. The current code therefore first constructs the dominant-mode flux from the standard multipolar flux formula,

$$\mathcal{F} = \frac{1}{8\pi} \sum_{\ell=2}^{\infty} \sum_{m=1}^{\ell} (m\Omega)^2 |h_{\ell m}|^2, \tag{59}$$

truncated to the $(2,2)$ mode only:

$$\mathcal{F}_{22} = \frac{1}{8\pi} (2\Omega)^2 |h_{22}|^2. \tag{60}$$

With the Newtonian factorization

$$h_{22} = h_{22}^N \hat{h}_{22}, \qquad h_{22}^N = -8\sqrt{\frac{\pi}{5}}\, \nu x\, e^{-2i\phi}, \tag{61}$$

this becomes

$$\mathcal{F}_{22}(x; \lambda_{\rm RG}) = \frac{32}{5} \nu^2 x^5 \left| \hat{h}_{22}(x; \lambda_{\rm RG}) \right|^2. \tag{62}$$

Since

$$\hat{h}_{22} = \hat{H}_{\rm eff}\, T_{22}\, \rho_{22}^2\, e^{i\delta_{22}}, \tag{63}$$

the current code is therefore using

$$\mathcal{F}_{22}(x; \lambda_{\rm RG}) = \frac{32}{5} \nu^2 x^5\, \hat{H}_{\rm eff}(x, \nu)^2\, |T_{22}(x, \nu; \lambda_{\rm RG})|^2\, \rho_{22}(x, \nu)^4. \tag{64}$$

The SPA phase is then built from the adiabatic balance law

$$\frac{dt}{dx} = -M\, \frac{d(E_{\rm real}/M)/dx}{\mathcal{F}_{22}(x)}, \tag{65}$$

with the circular-orbit relation $\Omega = x^{3/2}/M$. The Fourier-domain phase is assembled from the corresponding time-to-reference and gravitational-wave phase-to-reference integrals. The resulting default signal model for the $(2,2)$ mode is

$$\tilde{h}(f) = A_N(f) \sqrt{-\frac{2}{\nu} \frac{d(E_{\rm real}/M)}{dx}\bigg|_{x_f}}\, \exp\left[i \left(2\pi f t_c - \phi_c - \frac{\pi}{4} + \Psi_{22,\rm bal}(f; \lambda_{\rm RG}) + \arg \hat{h}_{22}(x_f) \right) \right], \tag{66}$$

with

$$x_f = (\pi M f)^{2/3}. \tag{67}$$

Here

$$A_N(f) = \sqrt{\frac{5}{24}}\, \frac{\mathcal{M}^{5/6}}{d_L\, \pi^{2/3}}\, f^{-7/6}. \tag{68}$$

So the current default $h(f)$ is:

restricted Newtonian SPA amplitude + conservative-energy SPA amplitude correction + 22-only balance-law SPA phase + complex Section IV phase $\arg \hat{h}_{22}(x_f)$.

This should be viewed as:

- paper-based in the construction of $\hat{h}_{22}$,
- more self-consistent than the older TaylorF2 wrapper because the phase is now driven by the same 22-mode flux that carries the RG deformation,
- still approximate because it uses only the $(2, 2)$ mode and a simplified detector-domain projection.

8.1 Amplitude options

The code keeps the older project proxy for diagnostics:

$$A_{\text{proxy}}(f; \lambda_{\text{RG}}) = A_N(f)\,|\hat{h}_{22}(x_f; \lambda_{\text{RG}})|. \tag{69}$$

With `amplitude_model="proxy"` and `phase_only=True`, this proxy amplitude is frozen to its GR value,

$$A_{\text{proxy,phase only}}(f) = A_N(f)\,|\hat{h}_{22}(x_f; \lambda_{\text{RG}} = 1)|, \tag{70}$$

while the phase still uses the requested λ_{RG}-dependent 22-only balance-law dynamics and the requested $\arg \hat{h}_{22}(x_f)$. For the default `amplitude_model="balance"`, the amplitude has no explicit λ_{RG}-dependent magnitude after the SPA cancellation, so `phase_only` does not change the default amplitude.

9 Practical Status

At the current point, the implementation is best described as:

- a clean Section IV-inspired $\hat{h}_{22}$ implementation,
- plus a practical 22-only balance-law SPA wrapper for Fisher work,
- with the test-mass branch retained for local validation.

It is suitable for the current standalone RG-tail Fisher studies, but it is not yet a full EOB waveform model and it is not integrated into the main population/network forecast pipeline in the repository.

10 Current Paper-Facing Figures

For the systems paper, the RG-tail material should stay compact. The current recommended figure set is:

- an expanded detector-and-network comparison plot of $\sigma(\lambda_{\text{RG}})$, generated from `results/rg_tail_forecast/s`
- a Gaussian Fisher corner plot for the CE 40 km forecast, showing the covariance among $\mathcal{M}_c$, η, d_L, and λ_{RG}.

The corresponding current files are:

- `results/rg_tail_forecast/summary/sigma_lambda_rg.pdf`,

- `results/rg_tail_forecast/summary/rg_tail_corner_ce40.pdf`.

The detector-and-network comparison figure is the cleanest headline output for the skill. In its current paper-facing version it includes:

$$\text{aLIGO}, \quad \text{Voyager}, \quad \text{CE 20 km}, \quad \text{CE 40 km}, \quad \text{ET}, \quad \text{CE 40+20 km}, \quad \text{ET + CE 40 km}, \quad \text{ET + CE 40+20 km} \tag{71}$$

For the current GW150914-like fiducial source with $f_{\max} = f_{\rm ISCO}$, the current comparison gives

$$\sigma(\lambda_{\rm RG}) \approx \begin{cases} 4.28 \times 10^{-1}, & \text{aLIGO}, \\ 2.14 \times 10^{-2}, & \text{Voyager}, \\ 3.58 \times 10^{-3}, & \text{CE 20 km}, \\ 5.29 \times 10^{-4}, & \text{CE 40 km}, \\ 6.00 \times 10^{-3}, & \text{ET}, \\ 5.02 \times 10^{-4}, & \text{CE 40+20 km}, \\ 5.25 \times 10^{-4}, & \text{ET + CE 40 km}, \\ 4.98 \times 10^{-4}, & \text{ET + CE 40+20 km}, \end{cases} \tag{72}$$

where the network points are reduced-network forecasts obtained by summing the independent Fisher matrices of the constituent detectors. In this reduced model, CE 40 km already carries most of the constraining power, so the network improvements are real but modest.

The CE corner plot is useful because it shows that the present $\lambda_{\rm RG}$ constraint is not only a one-number summary; it also reveals the dominant parameter degeneracies of the current skill-level model, especially the strong anticorrelation with $\mathcal{M}_c$ and the milder correlations with η and d_L.

11 Main Caveats

- The internal-tail factor in Eq. (131) is still omitted.
- The $\lambda_{\rm NS}^{\rm inst}$ denominator is not part of the current default tail implementation.
- The final $\tilde{h}(f)$ is still a repository-level stationary-phase construction; arXiv:2602.08833 provides the mode ingredients, not a complete detector-template prescription.
- The flux keeps only the $(2, 2)$ mode.
- The conservative dynamics are still the GR circular-orbit energy from Eq. (145); the RG deformation currently enters the radiative sector only, through the tail factor in $\hat{h}_{22}$.
- The network comparison is a reduced Fisher sum of constituent detector information and does not yet include sky location, polarization, or antenna-pattern geometry.
- The SPA amplitude is now derived from the same 22-only balance law, but the waveform still omits higher modes, spins, merger-ringdown, and detector response geometry.
- The whole RG-tail module remains a standalone research workflow rather than a production waveform package.

12 Bottom Line

The most concise current summary is:

> `waveform.py` now implements a clean Section IV external-tail $\hat{h}_{22}$, derives the corresponding 22-only flux $\mathcal{F}_{22} = (32/5)\nu^2 x^5 |\hat{h}_{22}|^2$, and uses that flux to build both the current Fourier-domain SPA phase and the default balance-law SPA amplitude for Fisher forecasts.

That is the current up-to-date description of the code.